

Lab 5: Multiple Linear Regression

Math 141, Week 5

Insert Your Name Here

Due: Tuesday, March 3rd at 11:59pm on Gradescope

In this lab, you will

1. Conduct exploratory data analysis (EDA) techniques in R with multiple predictors
2. Build linear regression models with multiple predictors
3. Interpret the estimated coefficients of multiple linear regression models
4. Build and interpret linear regression models with interactions
5. Assess how transformations can help satisfy LINE assumptions

Add any collaborators here!

I collaborated with...

Remember that you are not allowed to use any generative AI models (like Chat-GPT) on your lab.

```
#load the necessary packages
library(tidyverse)
library(pdxTrees)
library(moderndiver)
```

Exercises

Exercise 1: Regression with Transformation

Problems 1 & 2 use an extended version of the [Opportunity Insights dataset](#) (introduced in the activity from the *Data Wrangling* lecture) that has been joined with a dataset from The National Association of College and University Business Officers containing information on

university endowment assets from 2016.¹ The data are in the file `colleges_endow.Rdata`; for simplicity, the data frame is named `colleges`. The following table provides a list of the variables in the dataset and their descriptions. (Note: It is easier to read the table in the Rendered PDF document.)

Variable	Description
<code>name</code>	name of college
<code>region</code>	region, 1 for Northeast, 2 for Midwest, 3 for South, and 4 for West
<code>state</code>	state
<code>tier_name</code>	tier defined by selectivity and type
<code>type</code>	coded 1 for public, 2 for private non-profit, 3 for for-profit
<code>exp_instr_pc_2013</code>	instructional expenditures per student in 2013
<code>ipeds_enrollment_2013</code>	total undergraduate enrollment in Fall 2013
<code>sticker_price_2013</code>	average annual cost of attendance in 2013
<code>scorecard_netprice_2013</code>	net annual cost of attendance for bottom income quintile in 2013
<code>grad_rate_150_p_2013</code>	percentage of students graduating within 150% of normal time in 2013
<code>avgfacsal_2013</code>	average faculty salary in 2013
<code>sat_avg_2013</code>	average SAT scores (scaled to 1600) in 2013
<code>endowment_pc_2000</code>	endowment assets per student in 2000
<code>mr_kq5_pq1</code>	mobility rate, top 20% of the income distribution
<code>mr_ktop1_pq1</code>	mobility rate, top 1% of the income distribution
<code>par_median</code>	median parent household income
<code>par_q1</code>	fraction of parents in first (bottom) income quintile
<code>par_q2</code>	fraction of parents in second income quintile
<code>par_q3</code>	fraction of parents in third income quintile
<code>par_q4</code>	fraction of parents in fourth income quintile
<code>par_q5</code>	fraction of parents in fifth (top) income quintile
<code>par_top5pc</code>	fraction of parents in top 5% of income distribution
<code>par_top1pc</code>	fraction of parents in top 1% of income distribution
<code>k_median</code>	median child individual earnings in 2014 (at age 34)
<code>k_top5pc</code>	fraction of children in top 5% of income distribution
<code>k_top1pc</code>	fraction of children in top 1% of income distribution
<code>fy2016_endowment_pc</code>	endowment assests per student in 2016
<code>fy2016_endowment_thousands</code>	total value of endowment in thousands USD in 2016

Run the following R chunk to load the `colleges` data frame.

¹Specifically, the endowment data are from fiscal year 2016, which began on June 1, 2015 and ended on May 31, 2016. For the purposes of Math 141, it is fine to simply refer to these data as being collected in 2016.

```
load("data/colleges_endow.Rdata")
```

One metric of college access is net tuition price for low- and middle-income students after accounting for financial aid. The `scorecard_netprice_2013` variable contains data on the net annual cost of attendance for students with parental household income in the bottom quintile (in 2013); these data come from the U.S. Department of Education's College Scorecard.

A university's endowment refers to the financial assets invested by a university to support its educational and research mission; individuals and organizations can donate to contribute to a university's endowment. The `fy2016_endowment_pc` variable contains data on the monetary value of endowment assets per student (in 2016), measured in USD.

Let's investigate the association between college access and endowment assets by building a model for college access using endowment assets.

a. First, explore both the net tuition price variable and endowment assets variable individually. **Describe the distribution of each variable in the context of the data, referencing numerical and graphical summaries as needed.**

b. Construct a plot showing the association between net tuition price and endowment assets. Be sure to map the response variable to the y -axis and the explanatory variable to the x -axis.

c. Describe what the plot in part (b) shows; i.e., based on the plot, **does there seem to be a relationship between college access and endowment assets? Does a line seem like a reasonable summary for the relationship between these variables?** Explain your reasoning.

d. Add a variable to the `colleges` dataset that contains the log-transformed version of `fy2016_endowment_pc` and name it `log_endow_pc`. That is, if `fy2016_endowment_pc` equals \$1000, then the log-transformed version equals $\log(1000) = \ln(1000) = 6.908$. Statisticians tend to refer to natural log as just “log”. The function in R to compute the (natural) log is `log()`.

e. Recreate the plot from part (b), but use the log-transformed version of `fy2016_endowment_pc` rather than the original variable.

f. Based on the plot in part (e), does a line seem like a reasonable summary of the relationship between net tuition price and log-transformed endowment assets? If so, calculate the correlation coefficient and interpret the sign and magnitude.

g. Construct a simple linear regression model for net tuition price using the log-transformed endowment assets variable. **Print out the model coefficients table, and check the model diagnostics and comment on them.**

(Hint: Recall that you have the demo files from last lab with example code for building a linear regression model and creating model diagnostic plots.)

Exercise 2: Regression Models with Equal vs Varying Slopes

In this problem, we will investigate whether the relationship between net tuition price for low- and middle-income students and endowment value seems to differ for Ivy Plus schools versus other schools.

a. Create a new variable that categorizes whether or not a school is Ivy Plus. (*Hint: Revisit the code in slide 25 from 2/4*)

b. Construct a plot showing the association between net tuition price and log-transformed endowment assets; color the points based on whether a school is in the Ivy Plus. **Describe what you see. How does the average net tuition price for low- and middle-income students at Ivy Plus schools compare to other schools?**

(*Hint: Revisit the code in slide 7 from 2/23 for creating the plot, and slide 30 on 2/4 for calculating group averages.*)

c. Construct a linear regression model for net tuition price using log-transformed endowment assets *and* whether a school is categorized as Ivy Plus. *Assume equal slopes in the model.* Print the model coefficients table **and interpret the coefficients in the context of the data. Be sure to comment on whether the intercept makes sense in this context.**

d. In part (b), you estimated the difference in average net tuition price for low- to middle-income students at schools in the Ivy Plus versus schools not in the Ivy Plus. **Explain the reasoning behind why this difference is not equal to the coefficient for the Ivy Plus predictor variable in the model from part (c).**

e. Construct a linear regression model for the net tuition price using log-transformed endowment assets and whether a school is categorized as Ivy Plus. *Assume varying slopes in the model.* Based on this model, **write out the formula for the regression line for just Ivy Plus schools and the formula for the regression line for just non-Ivy Plus schools.** Using these formulas, state **what is the estimated slope for schools in the Ivy Plus and what is the estimated slope for non-Ivy Plus schools.**

f. Produce a plot that illustrates the linear model fit in part (e). Based on this plot and your previous analyses, is there evidence suggesting that making a donation to a school's endowment increases access for low- and middle-income students to the same extent at Ivy Plus schools as non-Ivy Plus schools? **Explain your reasoning.**

(Hint: Revisit the code in slides 19-20 from 2/23)

Exercise 3: Open-Ended Multiple Linear Regression

For this problem, we'll return to the `near_reed` dataset from Lab 04.

Your objective is simple: find a parsimonious, accurate, and sensible **multiple linear regression model to predict tree height**. As you embark on this mission, make sure to describe the steps you are taking in text along with the code you use to take those steps.

Once you've settled on a final model, report the following components: - Write out the full model formula and interpret two of its estimated coefficients in context. - Discuss the signs of the estimated regression coefficients. Are they what you expected them to be? - Show model diagnostic plots and assess them. - Discuss why your model is sensible for predicting tree height and justify your model building process.

Make sure to "show your work" in your model building process, i.e. keep all code used, plots made, and writing necessary to talk about what you did to come to your final model.

Some considerations:

- *You will want to try out several models. Make plots and summaries of predictor variables and the response variable to get a sense of what might be useful to use in your model. Remember that you can run `?get_pdxTrees_parks()` in the console to get information about the data.*
 - *Be careful to consider multicollinearity. Some predictors in the dataset are highly correlated with each other.*
 - *You might want to consider polynomial terms in your model, or transforming predictor variables to better fit the model.*
 - *You can use R^2 and adjusted R^2 to help you compare the models.*
-

```
# Grab trees near Reed
near_reed <- get_pdxTrees_parks() %>%
  filter(Park %in% c("Crystal Springs Rhododendron Garden",
                    "Kenilworth Park",
                    "Eastmoreland Garden",
                    "Berkeley Park"),
         !is.na(Functional_Type))

# Start your work here
```
